PROJECT PROGRESS REPORT

Delhi Air Quality Analysis & PM2.5 Prediction

Daily Technical Update | 4 November 2026

DATA PERIOD	SOURCE	EXTRACTED ROWS
January 2019	XKDR Forum API	148,747

Executive summary

Today's work focused on discovering, inspecting and extracting historical air-quality measurements for Delhi from the XKDR Forum India Air Quality dataset. The January 2019 national Parquet file was downloaded and queried with DuckDB. Delhi records for six selected pollutants were extracted to CSV.

The extracted CSV contains 148,747 measurement rows in long format. It is not yet a model-ready table: readings must first be aligned by monitoring station and timestamp.

1. Historical dataset discovery

• Identified monthly historical Parquet files exposed through the XKDR Forum API.

• Located and downloaded the January 2019 file to **data/raw/2019_01.parquet**.

• Verified an approximate file size of 2.05 MB; file metadata reports 1,056,000 national measurement records.

2. Schema inspection with DuckDB

Inspected the Parquet file with DuckDB rather than loading the full national dataset into Pandas.

Column	Meaning
station_id	Monitoring station identifier
station_name	Monitoring station name
state_name / city_name	Geographic metadata
parameter_name	Measured pollutant
unit	Measurement unit
collected_at	Measurement timestamp
value	Recorded measurement
source	Measurement source

3. Delhi coverage analysis

Filtering the January 2019 national dataset to Delhi returned 275,232 measurement records across 37 distinct stations. The station metadata list previously contained 40 Delhi stations; three listed stations had no records in this particular month.

The six selected pollutants and their January 2019 Delhi coverage are shown below.

Pollutant	Records	Stations
PM2.5	26,670	37
PM10	24,081	34
NO2	26,536	37
SO2	21,268	30
CO	24,373	37
Ozone	25,819	36

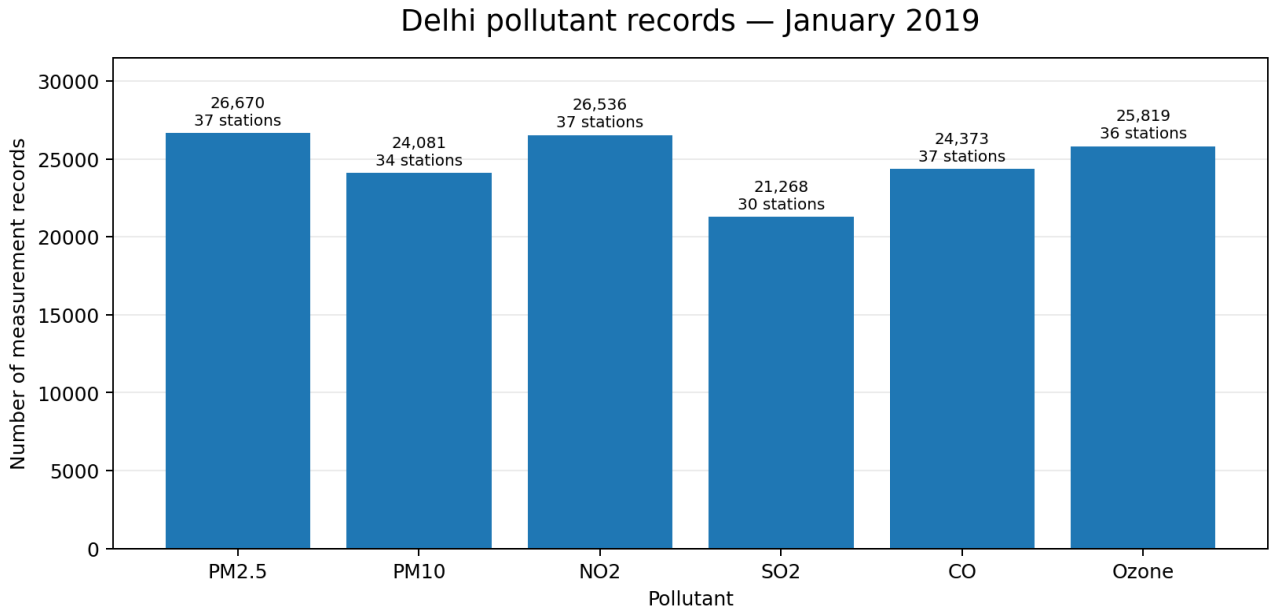


Figure 1. January 2019 pollutant record counts and station coverage in Delhi.

4. Delhi pollutant extraction

Created and executed **src/extract_delhi_data.py**. The script filters to **state_name = 'Delhi'** and the six selected pollutants: PM2.5, PM10, NO2, SO2 and CO, and Ozone.

Selected fields: **station_id, station_name, collected_at, parameter_name, value**.

Output: **data/processed/delhi_2019_01.csv**
Result: extraction completed successfully with **148,747 rows**.

5. Workflow status

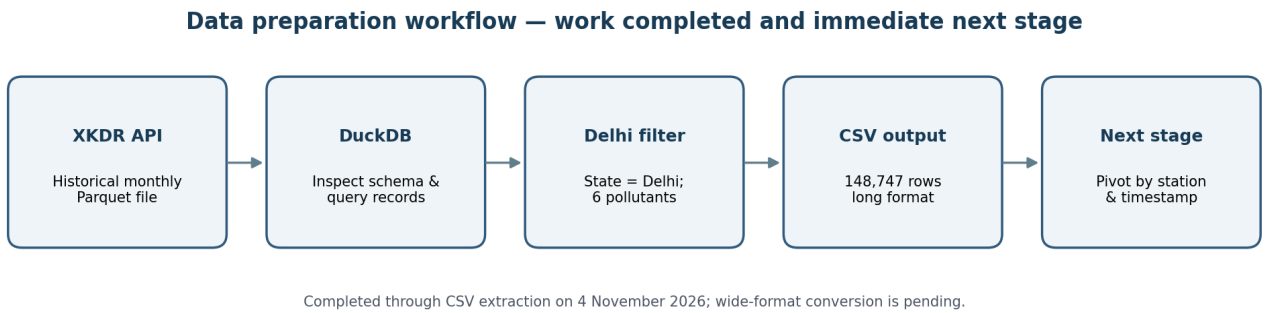


Figure 2. Data workflow completed today and the immediate next preparation stage.

6. Project decisions and data-quality considerations

• Target variable: PM2.5. Initial input features: PM10, NO2, SO2, CO and Ozone.

• Retain station_id and collected_at so measurements are aligned within the same station and timestamp.

• Do not automatically remove extreme values; inspect their context before classifying them as errors.

• Do not assume similarly named stations or stations at similar coordinates are interchangeable.

• Investigate missing measurements before choosing a row-filtering or imputation strategy.

• Keep credentials and local raw/processed datasets out of Git tracking; commit source code and documentation instead.

7. Git and repository update

A Git commit was created for the day's work. During push preparation, sensitive credential files were identified in the commit and removed from Git tracking, with ignore rules added. The corrected repository update was pushed successfully. Credentials should remain local and must never be committed.

8. Current status and next steps

Work item	Status
Historical file discovery and download	Completed
Parquet schema inspection with DuckDB	Completed
Delhi January 2019 coverage analysis	Completed
Six-pollutant extraction to CSV	Completed
Git cleanup and push	Completed
Convert long format to wide format	Next
Inspect missing values and data quality	Pending
Exploratory analysis and model preparation	Pending
Linear Regression training and evaluation	Pending

Next technical milestone

Reshape the long-format CSV into a wide-format table with one row per unique **(station_id, collected_at)** pair and separate columns for PM2.5, PM10, NO2, SO2, CO and Ozone. Then inspect row counts and missing-value patterns before deciding how to prepare the data for modelling.

Scope note: The current objective is contemporaneous PM2.5 estimation using other pollutant readings from the same station and timestamp. This is not a future forecasting model.

Data source: XKDR Forum India Air Quality API | Repository: delhi-air-quality-analysis-pm25-prediction